

Spacetime, propositional eternalism, and temporal egalitarianism

Handout for *Time and Modality* seminar

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1 Recap

- The primordial objection to the modal B-theory.
- The primordial objection to the temporal B-theory.
- The manifold picture.

2 Counterpart theory to the rescue?

Temporal counterpart theory (Sider, 'All the World's a Stage'). First pass:

- TCT a. To *have been* F is to have a past-counterpart which is F
b. To be *going to be* F is to have a future-counterpart which is F

... for any *qualitative* F . Modelled on Lewis's *modal* counterpart theory ('Counterpart theory and Quantified Modal Logic'). First pass on that:

MCT To be *possibly* F is to have a counterpart which is F

Key contrast between Lewis's modal counterpart theory and standard modal B-theory package: no hidden relativisation of predicates to worlds. There's a perfectly good question whether, e.g., Humphrey is an election-winner *simpliciter*. And there needs to be to make sense of MCT.

Similarly, temporal counterpart theory needs there to be a good question whether, e.g., Sider is sitting *simpliciter*.

3 Counterpart theories and bloated ontologies

4 Lewis's counterpart theory, carefully

Predicates:

Wx : x is a possible world. In Lewis's final theory: a maximal fusion of analogically spatiotemporally related objects. Call these *Lewis-worlds*.

Ixy : x is in y . In Lewis's final theory: either x is part of y , or x is a pure set (?).

Ax : x is actual. In Lewis's final theory: x is either analogically spatiotemporally related to me, or x is a pure set (?).

Cxy : x is a counterpart of y . In Lewis's final theory: x and y stand in a similarity relations. Which ones matter depends on the context we are in. (In some contexts, 'counterpart' may even stand for a relation that everything stands in to everything else!)

T1 The translation of ϕ is $[\phi]^@$ (ϕ holds in the actual world): that is, in primitive notation, $\exists w(\forall x(Ixw \equiv Ax) \& \phi^w)$.

Definition of ϕ^β :

T2a $[\phi]^w$ is ϕ , if ϕ is atomic.

T2b $[\neg\phi]^w$ is $\neg[\phi]^w$

T2c $[\phi \wedge \psi]^w$ is $[\phi]^w \wedge [\psi]^w$

T2g $[\forall x\phi]^w$ is $\forall x(Ixw \supset [\phi]^w)$.

T2h $[\exists x\phi]^w$ is $\exists x(Ixw \wedge [\phi]^w)$.

T2i $[\Box\phi(x_1, \dots, x_n)]^w$ is

$$\forall v\forall y_1 \dots \forall y_n((Wv \wedge Iy_1v \wedge Cy_1x_1 \wedge \dots \wedge Iy_nv \wedge Cy_nx_n) \rightarrow [\phi(y_1, \dots, y_n)]^v)$$

T2j $[\Diamond\phi(x_1, \dots, x_n)]^w$ is

$$\exists v\exists y_1 \dots \exists y_n(Wv \wedge Iy_1v \wedge Cy_1x_1 \wedge \dots \wedge Iy_nv \wedge Cy_nx_n \wedge [\phi(y_1, \dots, y_n)]^v)$$

Notes:

- Lewis uses a descriptive treatment of proper names. But let's bypass this issue by letting ourselves treat names as working in the same way as free variables.
- We need to assume that all the predicates in the language to which we are applying the translation are *qualitative*, otherwise we will get crazy results for things like 'Possibly, Merkel is a German cabaret singer'.
- We also need to exclude 'implicitly quantificational' predicates like 'invented bifocals' and 'shortest spy' from the language, otherwise we will get bad results for things like 'Necessarily, Frank is a shortest spy iff Frank is a spy and every spy is at least as tall as Frank'.
- Lewis later adopted a different (or at least more flexible) treatment of sentences with multiple names/free variables in the scope of a single modal operator.

A few examples:

Possibly, some swan is blue.

Possibly, every swan is blue.

Possibly, Obama is staring at a blue swan.

Possibly, Obama is staring at every blue swan.

Necessarily, every spy is a possible traitor.

5 Lewis's theory versus basic modal counterpart theory

Basic Modal Counterpart Theory (BMCT):

$$\Diamond\phi(x) := \exists y(Cyx \wedge \phi(y))$$

$$\Box\phi(x) := \forall y(Cyx \rightarrow \phi(y))$$

where everything except for 'x' in the formula $\phi(x)$ is qualitative.

Modal Redundancy: When ϕ is a closed sentence built up entirely out of qualitative vocabulary,

$\Box\phi$ and $\Diamond\phi$ are equivalent to ϕ .

Why BMCT makes it almost impossible to avoid *Modal Redundancy*. Choose some object a that has at least one counterpart, and consider the following instance of BMCT:

$$\Diamond(\phi \wedge (Fa \vee \neg Fa)) := \exists x(Cxa \wedge (\phi \wedge (Fx \vee \neg Fx)))$$